



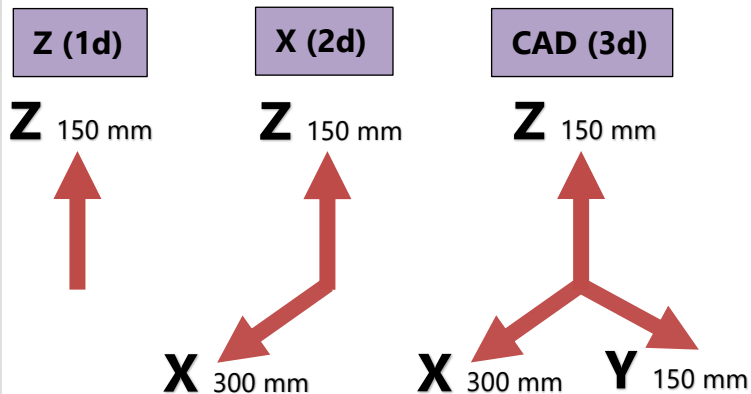
Technical Data Sheet TCM Eta

Machine with proven single-rotary piston system, fully powered by electric servo motors.

Wide range of applications.



Also available with housing



The picture above shows only one of the almost impossibly diverse variants that can be realized as a customer-oriented Eta.

Call us or contact us so we can bring your idea to a solution.

Single	9, 12	16, 18, 24	32, 36, 48	Optional For detailed information, see variants further down in the document
Boxer	9/18, 12/24	16/32, 18/36, 24/48	32/64, 36/72, 48/96	
Liner width	35 mm	45 mm	30 mm	
Piston distance	35 mm	22.5/45 mm	11 mm	
Piston stroke	52 mm	60 mm		
Conveyor length	3 rd Party Conveyor			
Mould widths	261 ... 412 mm			... 1'200 mm
Weight	2'000 kg	2'500 kg	3'000 kg	



Features

- ✓ Head movement by electric servo motors possible as single-axis (vertical: Z), two-axis (vertical and belt direction: X) or three-axis (CAD)
- ✓ High-precision single rotary piston system with identical dosing volumes (proven for more than 30 years – Single Piston System)
- ✓ No mixing between the pistons possible (1 piston = 1 nozzle)
- ✓ Machine designed for industrial application
- ✓ Quick changeover of different depositing tools (pistons/liners/nozzle plates) without additional tools
- ✓ Number of pistons individually selectable (easy activation / deactivation by using individual couplings)
- ✓ Easy cleaning of the dosing machine without additional tools (Stainless steel or aluminium in ALTEF® coating - for pH values 4 ... 8)
- ✓ Easy cleaning of the pistons and cylinder liners (no tools necessary)
- ✓ Movable touchscreen HMI for machine operation
- ✓ Depositing head in free-arm design for easy operation / cleaning / adjustment
- ✓ Fully digital machine control
- ✓ Remote access with PLC, HMI, servo drives possible
- ✓ Software designed for maximum accuracy and flexibility
- ✓ One Shot software already installed on every machine
- ✓ Double-walled hopper for the chocolate side

Options

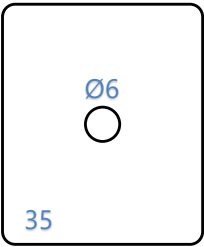
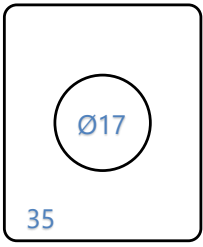
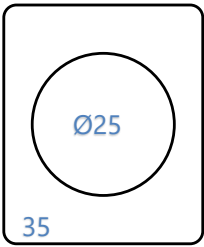
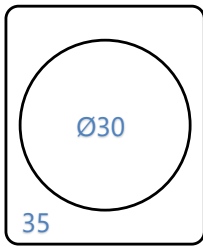
- For mould widths up to 1200 mm
- Double-walled hopper also for filling side
- Piston lubrication system for one or both piston sides (drip lubrication)
- Different agitator systems for one or both hoppers
- Triple Shot (1 filling / 2 shells OR 2 fillings / 1 shell)
- Quatro Shot (2 fillings / 2 shells)
- Control of downstream machine (e.g. vibration table) possible via control panel
- Y 300 mm (standard 150)
- RFID for mould recognition



Variants with 9 or 12 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X movement (belt running direction)
 - CAD for the highest flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot (OS) chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow shapes in widths of 261 ... 412 mm
 - For wide shapes in widths of 413 ... 1200 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Pistons/liners pairs in up to six different sizes (ø6 ... ø30 mm)

Piston Systems

Liner width 35 mm Piston stroke 52 mm				
Piston Ø	6 mm 8 mm 10 mm	17 mm	25 mm	30 mm
Volume	... 4.1 ml ... 4.9 g	11.8 ml 14.2 g	25.5 ml 30.6 g	36.8 ml 44.1 g
Recommendation	Decorations or drops	Chocolates in One Shot, Triple Shot or Quatro Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine capacities (examples at 12 pistons per side, thus 24 pistons in total if as a boxer)

Example 1: 100g bars without filling

Maximum design with two chocolate hoppers (3 nozzles per bar)
 Per stroke 8 panels, 17 strokes/min and 60 min/h yields:

- **8,160 bars/h or 816 kg/h**

Example 2: 10g One Shot Chocolates

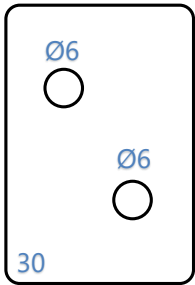
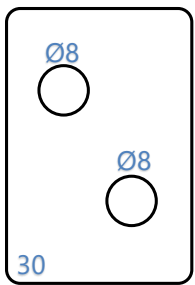
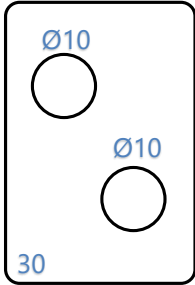
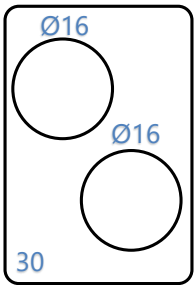
Maximum design with two hoppers (chocolate and filling)
 Each stroke 12 chocolates, 20 strokes/min and 60 min/h results in:

- **14,400 chocolates/h or 144 kg/h**

Variants with 32, 36 or 48 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops without filling)
 - Boxers (for one-shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow moulds in widths of 261 ... 412 mm
 - For wide moulds in widths of 413 ... 1200 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to four different sizes (Ø6 ... Ø16 mm)

Piston Systems

Liner width 30 mm Piston stroke 60 mm				
Piston Ø	6 mm	8 mm	10 mm	16 mm
Volume	1.7 ml	3.0 ml	4.7 ml	12.1 ml
	2.0 g	3.6 g	5.7 g	14.5 g
Recommendation	Decorations very fine	Decorations	Decorations coarse	Chocolates in One Shot, Triple Shot or Quatro Shot

Machine capacities (examples at 72 pistons per side, thus 144 pistons in total if as a boxer)

Example 1: 100g bars without filling (solid)

- Possible but machine not suitable for blackboards

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

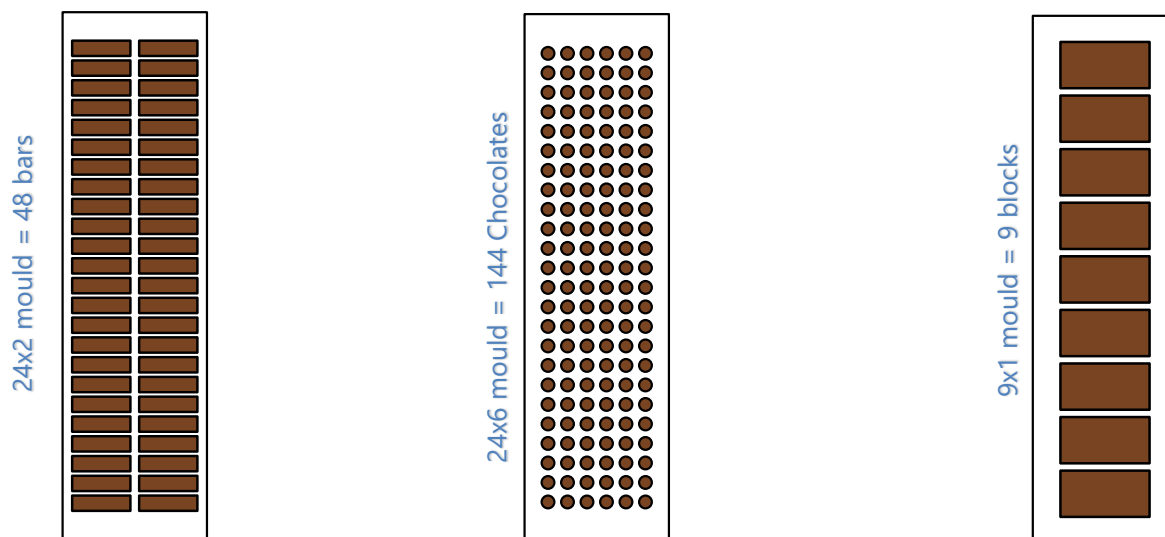
Each stroke 72 chocolates, 20 strokes/min and 60 min/h results in:

- **86,400 chocolates/h or 864 kg/h**

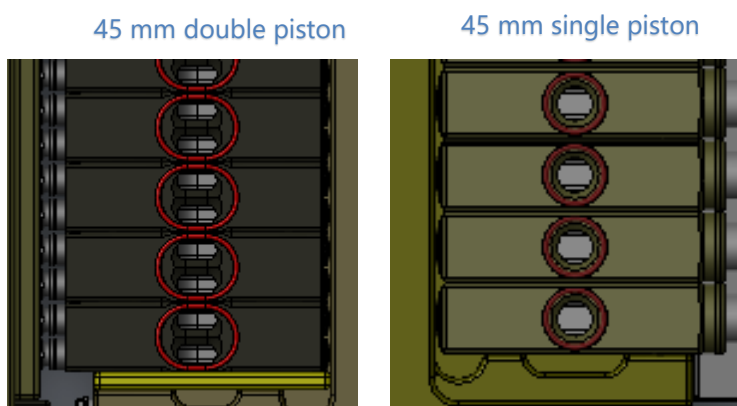


Combination options

With a liner width of 45 mm, the system is equipped with double pistons. Thus, for example, a single piston per liners can be equipped individually on the left side and double-piston liner on the right side. This increases variability. Here are examples with moulds in dimension 830 mm x 230 mm:



For the alignment of the off-centred pistons, dummies can serve as placeholders.

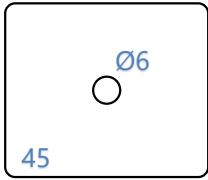
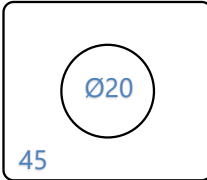
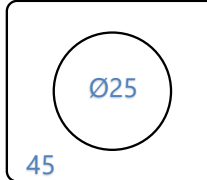
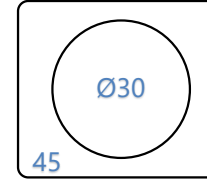


Piston ø mm	Stroke 52 mm 35	Piston volume		Stroke 60 mm			Piston volume	
		ml	g	30/2	45/2	45	ml	g
6	X	1.47	1.76	X	X	X	1.70	2.04
8	X	2.61	3.14	X	X	X	3.02	3.62
10	X	4.08	4.90	X	X	X	4.71	5.65
16				X			12.06	14.48
17	X	11.80	14.16					
20					X	X	18.85	22.62
25	X	25.53	30.63			X	29.45	35.34
30	X	36.76	44.11			X	42.41	50.89
40						X	75.40	90.48

Variants with 9 or 12 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for one-shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow moulds in widths of 261 ... 412 mm
 - For wide moulds in widths of 413 ... 1200 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to six different sizes (Ø6 ... Ø30 mm)

Piston Systems

Liner width 45 mm Piston stroke 60 mm				
Piston Ø	6 mm 8 mm 10 mm	20 mm	25 mm	30 mm
Volume	... 4.7 ml ... 5.7 g	18.9 ml 22.6 g	29.5 ml 35.3 g	42.4 ml 50.9 g
Recommendations	Lids or drops	Chocolates in One Shot, Triple Shot or Quatro Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine Capacities (Examples with 12 pistons each side, 24 pistons in total, if as Boxer)

Example 1: 100g Bars without filling (solid)

Maximum design with two chocolate hoppers (2 nozzles for one bar)

Each stroke 12 bars, 15 strokes/min and 60min/h results in:

- **10'800 bars/h or 1'080 kg/h**

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

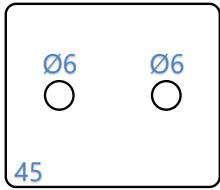
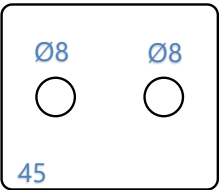
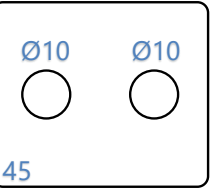
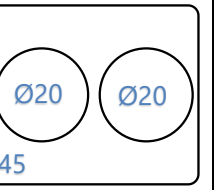
Each stroke 12 chocolates, 20 strokes/min and 60min/h results in:

- **14'400 chocolates/h or 144 kg/h**

Variants with 16, 18 or 24 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxers (for one-shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow moulds in widths of 261 ... 412 mm
 - For wide moulds in widths of 413 ... 1200 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to four different sizes (Ø6 ... Ø20 mm)

Piston Systems

Liner width 45 mm Piston stroke 60 mm				
Piston Ø	6 mm	8 mm	10 mm	20 mm
Volume	1.7 ml	3.0 ml	4.7 ml	18.9 ml
	2.0 g	3.6 g	5.7 g	22.6 g
Recommendation	Decorations very fine	Decorations	Decorations coarse	Chocolates in One Shot, Triple Shot or Quatro Shot

Machine capacities (examples at 24 pistons per side, thus 48 pistons in total if as a boxer)

Example 1: 100g bars without filling (solid)

- Possible but machine not ideal for bars

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

Each stroke 24 chocolates, 20 strokes/min and 60 min/h results in:

- **28,800 chocolates/h or 288 kg/h**



Applications



- Bars without filling
- Bars with small inclusions up to 3 mm
- Bars with whole nuts



- Filled chocolates in 2 colours (One Shot)
- Filled candy bars in 2 colours (One Shot)
- Filled chocolates in 3 colours (Triple Shot)
- Filled chocolates in 4 colours (Quatro Shot)
- Filled chocolate bars (One Shot Curtain)



- Free forms without filling
- Free forms with fillings
- Free forms with two fillings



- Fillings for Macarons
- Fillings for Alfajores



- Decorations small 0.1g (eyes)
- Decorations big (body)

